Sai Venkat Gunda

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SUMMARY

Advanced Controls and ML Engineer with 4 years of industry experience. Specialized in **Reinforcement Learning, Modern Control and Machine Learning.** High agency. Always curious. Currently targeting roles in the **Physical AI** space that require a deep mix of physical modeling, simulation, and applied ML.

Skills

Languages: Python, C++ **Modelling and Simulation**: MATLAB, Simulink, Gazebo, MuJoCo

ML & Robotics: ROS2(Kilted Kaiju), LeRobot, stable-baselines3, PyTorch, Scikit-learn, Reinforcement Learning(PPO, SAC, DQN), Imitation Learning(ACT, Diffusion Models)

Professional Experience

- **Jaguar Land Rover** — Bengaluru, India
  Advanced Controls & ML Engineer — *May 2024 - Present*
  - **Virtual Sensors**: Engineered and deployed industrialized ML models to virtualize thermal sensors achieving **96%** real time **prediction accuracy** in production vehicles.
  - **ML Data Pipelines**: Built an end-to-end ML pipeline to create Virtual Sensors starting from data collection, pre-processing, filtering, modelling and validation. Reduced the complete delivery cycle from **1 week to 4 hours.**
  - **Engineering Lifecycle**: Managed the complete delivery of Hypervisor feature enabling a **range benefit of 10+ miles** from planning, prototyping, industrialization, validation and delivery.
  - **State Estimation(EKF)**: Developing an Extended Kalman Filter for the refrigerant system to replace existing Virtual Sensors with grey-box state estimation techniques for better explainability and reliability of predictions.
  - **Model Predictive Control(MPC)**: Currently developing a deployable Model Predictive Controller architecture for vehicle refrigerant systems to optimize between comfort, performance, component durability and efficiency.
- **Jaguar Land Rover** — Bengaluru, India
  Structural CAE Analyst — *July 2022 - May 2024*
  - **Reduced Order Modelling**: Developed reduced order 1D frontal crash modelling techniques providing quick verification of design ideas in seconds with an **accuracy of 80%**.

Projects

- **Intrinsic AI for Industry Challenge** — Bengaluru, India
  Robotics challenge by Intrinsic, Google — *Apr 2026 - Present*
  - **Robot Learning Pipeline**: Built an end-to-end **robot learning pipeline** in Gazebo, generating 1,000+ training trajectories with kinematic perturbations and domain randomization to ensure robust policy training.
  - **ACT Policy**: Fine-tuned an Action Chunking with Transformers (ACT) policy using a **ResNet18** backbone via **LeRobot**; achieved **sub-10mm positioning accuracy** but identified failure modes during contact-rich insertion.
  - **Diffusion Policy**: Implementing a diffusion-based control policy to specifically resolve ACT's contact dynamic failures, targeting **high-precision, sub-millimeter port insertion**.
- **Autonomous All-terrain Mars Rover** — IIT Madras
  Team and Traversal Lead — *Jan 2018 - May 2021*
  - **5 DOF Manipulator**: Developed a **5 DOF serial manipulator** capable of lifting **5 kg** with a compliant gripper. Developed closed loop control systems for the manipulator using Inverse Kinematics for robust manipulation.
  - **All-Terrain Traversal System**: Developed a rocker bogie suspension system for all terrain travel of autonomous rovers capable of climbing **45 cm heights and 30° slopes**.

Education

- **Indian Institute of Technology Madras** — Chennai, India
  Dual Degree(B.Tech + M.Tech) Honours in Mechanical Engineering; GPA: 9.14 — *Aug. 2017 – May. 2022*